

Magnetostatics, Faraday's Law, Quasistatic Fields

PHYS505 – Electromagnetic Theory

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Magnetostatics

- *Magnetostatics* means no change in net charge density

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0, \quad \frac{\partial \rho}{\partial t} = \vec{\nabla} \cdot \vec{J} = 0$$

- $\vec{B}$ referred to as magnetic flux density or magnetic induction
- Sources characterized by current density $\vec{J}$, or current I for a thin wire
- Connection between the two expressed by Biot-Savart Law

$$d\vec{B}(\vec{x}) = \frac{\mu_0}{4\pi} I \frac{d\vec{l}' \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3}$$

- $1/r^2$ in magnitude like $\vec{E}$ -field from ρ but the vector form is completely different

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Differential equations for magnetostatics

- Similar to the expression for $\vec{E}$ from ρ in electrostatics

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{x}') \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} d^3x' = -\frac{1}{4\pi\epsilon_0} \vec{\nabla} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

which explicitly shows $\vec{\nabla} \times \vec{E} = \vec{0}$

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Ampère's Law

- The curl equation

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

- integrated over a surface S bounded by curve C , gives

$$\int_S \vec{\nabla} \times \vec{B} \cdot \hat{n} \, da = \mu_0 \int \vec{J} \cdot \hat{n} \, da = \mu_0 I$$

- From Stokes' theorem

$$\int_S \vec{\nabla} \times \vec{B} \cdot \hat{n} \, da = \int_C \vec{B} \cdot d\vec{l}$$

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Scalar potential

- Magnetostatics involves solving for $\vec{B}$ in the differential equations

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}, \quad \vec{\nabla} \cdot \vec{B} = 0$$

- In regions with no current density,

$$\vec{\nabla} \times \vec{B} = \vec{0}$$

- $\vec{B}$ can be expressed as the gradient of a *magnetic scalar potential* Φ_M

$$\vec{B} = -\vec{\nabla}\Phi_M$$

- so we end up with the Laplace eqn

$$\nabla^2 \Phi_M = 0$$

- and can apply the techniques we know from electrostatics

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- The divergence equation means $\vec{B}$ can be expressed as the curl of a *vector potential* $\vec{A}$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

- We already have part of the answer

$$\vec{B}(\vec{x}) = \frac{\mu_0}{4\pi} \int \vec{J}(\vec{x}') \times \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} d^3x' = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

- where we can identify

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x' + \vec{\nabla} \psi(\vec{x})$$

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- The transformation

$$\vec{A} \rightarrow \vec{A} + \vec{\nabla}\psi$$

where ψ is an arbitrary (differentiable) scalar function, is called a *gauge transformation*

- and leaves $\vec{B}$ invariant
- Substituting $\vec{A}$ into the curl eqn for $\vec{B}$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

- If we choose the *Coulomb gauge*,

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Sidebar on symmetry

- In the quantum version of the theory (QED), gauge invariance implies conservation of charge
- Example of general principle connecting symmetry with conservation laws
- Formally expressed in Noether's theorem: there is a conserved "current" associated with every differentiable symmetry of the action
- One of the most fundamental ideas in modern physics
- Other examples: space/time translation, rotation



Invariante Variationsprobleme.

(F. Klein zum fünfzigjährigen Doktorjubiläum.)

Yan

Emmy Noether in Göttingen.

Vorgelegt von F. Klein in der Sitzung vom 26. Juli 1918/9.

Es handelt sich um Variationsprobleme, die eine kontinuierliche Gruppe (im Lieischen Sinne) gestatten; die daraus sich ergebenden Folgerungen für die zugehörigen Differentialgleichungen finden ihren allgemeinsten Ausdruck in den in § 1 formulierten, in den folgenden Paragraphen bewiesenen Sätzen. Über diese aus Variationsproblemen entpringenden Differentialgleichungen lassen sich dann wiederum Aussagen gewinnen, die wiederum zur Gruppe gestattender Differentialgleichungen, die den Gegenstand der Lieischen Untersuchungen bilden. Das folgende beruht also auf einer Verbindung der Methoden der formalen Variationsrechnung mit denen des Lieischen Gruppentheorie. Für spezielle Gruppen und Variationsprobleme ist diese Verbindung der Methoden nicht neu; ich erwähne Hamel und Herglotze für spezielle endliche, Lorents und Noether für die Gruppe der $(n+1)$ -Felder, Weyl und Noether für die Gruppe der $(n+1)$ -Felder. Immerhin ist die zweite, kleinere Note und die verliegenden Ausführungen gewissermaßen durch einander beeinflusst.

Sidebar on symmetry

- In the quantum version of the theory (QED), gauge invariance implies conservation of charge
- Example of general principle connecting symmetry with conservation laws
- Formally expressed in Noether's theorem: there is a conserved "current" associated with every differentiable symmetry of the action
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Vorgelegt von F. Klein in der Sitzung vom 28. Juli 1918⁹⁾

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Vector potential

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x' + \vec{\nabla}\psi(\vec{x})$$

- Coulomb gauge means

$$\vec{\nabla} \cdot \vec{A} = \frac{\mu_0}{4\pi} \int \vec{\nabla} \cdot \left(\frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} \right) d^3x' + \nabla^2\psi(\vec{x})$$

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Circular current loop

- in the $x - y$ plane with radius a and current I

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$$\vec{J}(r', \theta', \phi') = I \sin \theta' \delta(\cos \theta') \frac{\delta(r' - a)}{a} \hat{\phi}'$$

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- vector potential for $\vec{x}$ in $x - z$ plane

$$\begin{aligned} A_{\phi}(r, \theta) &= \frac{\mu_0 I}{4\pi a} \int dr' r'^2 d\Omega \frac{\sin \theta' \cos \phi' \delta(r' - a)}{|\vec{x} - \vec{x}'|} \\ &= \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{\cos \phi' d\phi'}{\sqrt{a^2 + r^2 - 2ar \sin \theta \cos \phi'}} \\ &= \frac{\mu_0}{4\pi} \frac{4Ia}{\sqrt{a^2 + r^2 - 2ar \sin \theta \cos \phi'}} \left[\left(\frac{2}{k^2} - 1 \right) K(k) - \frac{2}{k^2} E(k) \right] \\ k^2 &= \frac{4ar \sin \theta}{a^2 + r^2 + 2ar \sin \theta} \end{aligned}$$

where $K(k)$, $E(k)$ are elliptic integrals

Circular current loop

- in the $x - y$ plane with radius a and current I
- corresponding $\vec{B}$

$$B_r = \frac{\partial_\theta (\sin \theta A_\phi)}{r \sin \theta}, \quad B_\theta = -\frac{\partial_r (r A_\phi)}{r}, \quad B_\phi = 0$$

Circular current loop

- in the $x - y$ plane with radius a and current I
- Approximate expansion gives

$$A_\phi = \frac{\mu_0 I a^2 r \sin \theta}{4 (a^2 + r^2)^{3/2}} \left[1 + \frac{15}{8} \frac{a^2 r^2 \sin^2 \theta}{(a^2 + r^2)^2} + \dots \right]$$

$$B_r = \frac{\mu_0 I a^2 \cos \theta}{2 (a^2 + r^2)^{3/2}} \left[1 + \frac{15}{4} \frac{a^2 r^2 \sin^2 \theta}{(a^2 + r^2)^2} + \dots \right]$$

$$B_\theta = -\frac{\mu_0 I a^2 \sin \theta}{4 (a^2 + r^2)^{5/2}} \left[2a^2 - r^2 + \frac{15}{8} \frac{a^2 r^2 \sin^2 \theta (4a^2 - 3r^2)}{(a^2 + r^2)^2} + \dots \right]$$

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- Far from the loop, $r \gg a$

$$B_r \simeq \frac{\mu_0}{2\pi} \left(I \pi a^2 \right) \frac{\cos \theta}{r^3}, \quad B_\theta \simeq \frac{\mu_0}{4\pi} \left(I \pi a^2 \right) \frac{\sin \theta}{r^3}$$

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- magnetic dipole moment

$$m = I \pi a^2$$

Circular current loop

- in the $x - y$ plane with radius a and current I
- General expansion in spherical harmonics

$$A_\phi = -2 \frac{\mu_0 I a}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (2n-1)!!}{2^n (n+1)!} \frac{r_{<}^{2n+1}}{r_{>}^{2n+2}} P_{2n+1}^1(\cos \theta)$$

$$B_r = -\frac{\mu_0 I a}{2r} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{r_{<}^{2n+1}}{r_{>}^{2n+2}} P_{2n+1}^1(\cos \theta)$$

$$B_\theta = -\frac{\mu_0 I a^2}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n (n+1)!} P_{2n+1}^1(\cos \theta) \\ \times \begin{cases} -\frac{2n+2}{2n+1} \frac{1}{a^3} \left(\frac{r}{a}\right)^{2n}, & r < a \\ \frac{1}{r^3} \left(\frac{a}{r}\right)^{2n}, & r > a \end{cases}$$

Field from localized current distribution

- Far from the source ($|\vec{x}| \gg |\vec{x}'|$)

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{1}{|\vec{x}|} + \frac{\vec{x} \cdot \vec{x}'}{|\vec{x}|^3} + \dots$$

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- Taylor expansion for components of $\vec{A}$

$$A_i(\vec{x}) = \frac{\mu_0}{4\pi} \left[\frac{1}{|\vec{x}|} \int J_i(\vec{x}') d^3x' + \frac{\vec{x}}{|\vec{x}|^3} \cdot \int J_i(\vec{x}') \vec{x}' d^3x' + \dots \right]$$
$$\vec{A} = \frac{\mu_0}{4\pi} \left\{ \frac{1}{|\vec{x}|} \int \vec{J}(\vec{x}') d^3x' - \frac{1}{2} \frac{1}{|\vec{x}|^3} \vec{x} \times \int [\vec{x}' \times \vec{J}(\vec{x}')] d^3x' + \dots \right\}$$

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$$\mathcal{M}(\vec{x}) = \frac{1}{2} \vec{x} \times \vec{J}(\vec{x})$$

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- For planar current loop

$$\vec{m} = \frac{I}{2} \oint \vec{x} \times d\vec{l} = I \hat{n} \int da$$

Current due to moving point charges

- Point charges q_i of mass M_i moving with velocity $\vec{v}_i$

$$\vec{J} = \sum_i q_i \vec{v}_i \delta(\vec{x} - \vec{x}_i)$$

- Magnetic moment

$$\vec{m} = \frac{1}{2} \sum_i q_i (\vec{x}_i \times \vec{v}_i) = \sum_i \frac{q_i}{2M_i} \vec{L}_i$$

- If all the particles have the same q_i/M_i

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Force, torque, energy

- Current distribution in external $\vec{B}$

$$B_k(\vec{x}) = B_k(\vec{0}) + \vec{x} \cdot \vec{\nabla} B_k(\vec{0}) + \dots$$

- experiences a force

$$\begin{aligned}\vec{F} &= \int \vec{J}(\vec{x}) \times \vec{B}(\vec{x}) d^3x \\ &= (\vec{m} \times \vec{\nabla}) \times \vec{B} + \dots \simeq \vec{\nabla}(\vec{m} \cdot \vec{B}) - \vec{m}(\vec{\nabla} \cdot \vec{B}) = \vec{\nabla}(\vec{m} \cdot \vec{B})\end{aligned}$$

- and a torque

$$\begin{aligned}\vec{N} &= \int \vec{x} \times [\vec{J}(\vec{x}) \times \vec{B}(\vec{x})] d^3x \\ &= \int \vec{x}' \times [\vec{J}(\vec{x}') \times \vec{B}(\vec{0})] d^3x' + \dots \simeq \vec{m} \times \vec{B}(\vec{0})\end{aligned}$$

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$$U = -\vec{m} \cdot \vec{B}$$

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- For ferromagnetic materials

$$\vec{B} = \vec{B}[\vec{H}]$$

- Boundary between media with surface normal $\hat{\mathbf{n}}$ pointing from region 1 to 2

$$\hat{\mathbf{n}} \cdot (\vec{B}_2 - \vec{B}_1) = 0$$
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- Need to solve

$$\vec{\nabla} \cdot \vec{B} = 0, \quad \vec{\nabla} \times \vec{H} = \vec{J}$$

subject to constitutive relation between $\vec{B}$ and $\vec{H}$

- Simplifies for linear media

$$\vec{\nabla} \times \left(\frac{1}{\mu} \vec{\nabla} \times \vec{A} \right) = \vec{J}$$

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- For linear media

$$\vec{\nabla} \cdot (\mu \vec{\nabla}\Phi_M) = 0$$

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BVP with hard ferromagnets ($\vec{J} = \vec{0}$, $\vec{M} \neq \vec{0}$)

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with effective magnetic-charge density ρ_M

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Uniformly magnetized sphere

- Fixed $\vec{M} = M_0 \hat{\mathbf{z}}$ and effective $\sigma_M = \hat{\mathbf{n}} \cdot \vec{M} = M_0 \cos \theta$

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- and the expansion for $1/|\vec{x} - \vec{x}'|$, only the $l = 1$ term is non-0

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- Outside, it's a pure dipole field

$$\Phi_M(r > a, \theta) = \frac{1}{3} M_0 a^3 \frac{\cos \theta}{r^2} = \frac{\hat{r} \cdot \vec{m}}{4\pi r^2}, \quad \vec{m} = \frac{4\pi}{3} a^3 \vec{M}$$

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Magnetized sphere in external field

- Uniformly magnetized sphere in external field $\vec{H}_0$
- Superposition – add $\vec{B}_0$ to solution

$$\vec{H}_{in} = \frac{1}{\mu_0} \vec{B}_0 - \frac{1}{3} \vec{M}, \quad \vec{B}_{in} = \vec{B}_0 + \frac{2}{3} \mu_0 \vec{M}$$

- If the sphere is para/diamagnetic with permeability μ

$$\vec{B}_{in} = \mu \vec{H}_{in}, \quad \vec{B}_0 + \frac{2}{3} \mu_0 \vec{M} = \mu \left(\frac{1}{\mu_0} \vec{B}_0 - \frac{1}{3} \vec{M} \right)$$

- which gives

$$\vec{M} = \frac{3}{\mu_0} \frac{\mu - \mu_0}{\mu + 2\mu_0} \vec{B}_0$$

- If the sphere is ferromagnetic, we use the *hysteresis* curve for $\vec{B}_{in}$ vs $\vec{H}_{in}$ together with

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Magnetic shielding

- Spherical shell of inner/outer radius a/b with permeability μ placed in region with uniform $\vec{H}_0$
- $\vec{J} = \vec{0}$, so we can use the scalar potential for the solution to Laplace eqn

$$\Phi_M = -H_0 r \cos \theta + \sum_{l=0}^{\infty} \frac{\alpha_l}{r^{l+1}} P_l(\cos \theta), \quad r > b$$

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$$\Phi_M = \sum_{l=0}^{\infty} \delta_l r^l P_l(\cos \theta), \quad r < a$$

- Matching the solutions, $\alpha_l = \beta_l = \gamma_l = 0, \quad l \neq 1$
- The field outside is $\vec{H}_0$ plus that of a $\parallel$ dipole moment α_1
- The field inside is uniform, parallel to $\vec{H}_0$ and $\propto 1/\mu$

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- The field inside is uniform, parallel to $\vec{H}_0$ and $\propto 1/\mu$

Magnetic shielding

- Spherical shell of inner/outer radius a/b with permeability μ placed in region with uniform $\vec{H}_0$
- $\vec{J} = \vec{0}$, so we can use the scalar potential for the solution to Laplace eqn

$$\Phi_M = -H_0 r \cos \theta + \sum_{l=0}^{\infty} \frac{\alpha_l}{r^{l+1}} P_l(\cos \theta), \quad r > b$$

$$\Phi_M = \sum_{l=0}^{\infty} \left(\beta_l r^l + \frac{\gamma_l}{r^{l+1}} \right) P_l(\cos \theta), \quad a < r < b$$

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Faraday's law of induction

- Magnetic flux linking circuit C bounded by surface S

$$F = \int_S \vec{B} \cdot \hat{n} da$$

- *Electromotive force* around the circuit

$$\mathcal{E} = \oint_C \vec{E}' \cdot d\vec{l}$$

- Faraday's law relates the two

$$\mathcal{E} = -k \frac{dF}{dt}$$

- where the -sign is meant to indicate that the induced current opposes the change (Lenz's law)
- The constant k follows from Galilean invariance ($k = 1$ in SI units)

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- *Convective derivative* for a vector field $\vec{B}$ in moving media

$$\frac{d}{dt}\vec{B} = \partial_t\vec{B} + \sum_k \left(\frac{\partial}{\partial x_k}\vec{B} \right) \left(\frac{d}{dt}x_k \right) = \partial_t\vec{B} + (\vec{v} \cdot \vec{\nabla}) \vec{B}$$
$$(\vec{v} \cdot \vec{\nabla}) \vec{B} = \vec{\nabla} \times (\vec{B} \times \vec{v}) + \vec{v} (\vec{\nabla} \cdot \vec{B})$$

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$$\oint_C [\vec{E}' - k(\vec{v} \times \vec{B})] \cdot d\vec{l} = -k \int_S \partial_t \vec{B} \cdot \hat{n} da$$

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- Differential form follows from Stokes' theorem

$$\vec{\nabla} \times \vec{E} + \partial_t \vec{B} = 0$$

Energy in magnetic field

- Need to consider induced emf when calculating the total energy required to assemble a system of steady state currents and magnetic fields
- Charge q moving with velocity $\vec{v}$ against force $q\vec{E}$ requires work at a rate of $q\vec{v} \cdot \vec{E}$
- Total for all the charges moving within the circuit

$$\frac{dW}{dt} = - \oint_C dq \vec{v} \cdot \vec{E} = - \oint_C dq \frac{d\vec{l}}{dt} \cdot \vec{E} = -I\mathcal{E} = I \frac{dF}{dt}$$

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- If $\vec{J}$ is linear in $\vec{A}$,

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- Placing an object with permeability μ_1 in a configuration with fixed sources and $\vec{B}_0, \mu_0$ leads to

$$\begin{aligned} W &= \frac{1}{2} \int_{V_1} (\vec{B} \cdot \vec{H}_0 - \vec{H} \cdot \vec{B}_0) d^3x \\ &= \frac{1}{2} \int_{V_1} (\mu_1 - \mu_0) \vec{H} \cdot \vec{H}_0 d^3x \\ &= \frac{1}{2} \int_{V_1} \left(\frac{1}{\mu_0} - \frac{1}{\mu_1} \right) \vec{B} \cdot \vec{B}_0 d^3x \end{aligned}$$

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- If the medium is free space

$$W = \frac{1}{2} \int_{V_1} \vec{M} \cdot \vec{B}_0 d^3x$$

- System with N circuits in empty space

$$W = \frac{1}{2} \sum_{i=1}^N L_i I_i^2 + \sum_{i=1}^N \sum_{j>i}^N M_{ij} I_i I_j$$

- in terms of *self inductance* L_i and *mutual inductance* M_{ij}

$$W = \frac{\mu_0}{8\pi} \int d^3x \int d^3x' \frac{\vec{J}(\vec{x}) \cdot \vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|}$$

$$L_i = \frac{\mu_0}{4\pi I_i^2} \int_{C_i} d^3x_i \int_{C_i} d^3x'_i \frac{\vec{J}(\vec{x}_i) \cdot \vec{J}(\vec{x}'_i)}{|\vec{x}_i - \vec{x}'_i|}$$

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Quasi-static magnetic fields in conductors

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- Conducting media with Ohm's law relating $\vec{J}$ and $\vec{E}$

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- which, in the Coulomb gauge, gives a *diffusion eqn* for $\vec{A}$

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- and for $\vec{E}$, using $\vec{\nabla} \cdot \vec{E} = 0$

$$\begin{aligned} \vec{\nabla} \times \vec{\nabla} \times \vec{E} &= \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} \\ &= -\vec{\nabla} \times \partial_t \vec{B} = -\mu \partial_t \vec{J} \\ \nabla^2 \vec{E} &= \mu \sigma \partial_t \vec{E} \end{aligned}$$

Diffusion of fields in conductors

- From the equations for $\vec{A}, \vec{E}$

$$\nabla^2 \vec{E} = \mu\sigma \partial_t \vec{E}$$

- order-of-magnitude estimate of decay time τ for fields over length scale L

$$\tau = \mathcal{O}(\mu\sigma L^2)$$

- Penetration depth δ for external field with frequency $\nu = 1/\tau$

$$\delta = \mathcal{O}\left(\frac{1}{\sqrt{\mu\sigma\nu}}\right)$$

- For copper, $\mu\sigma \simeq 75 \text{ s/m}^2 = 75 / (\text{mm}^2\text{-MHz})$,
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Diffusion of fields in conductors

- From the equations for $\vec{A}, \vec{E}$

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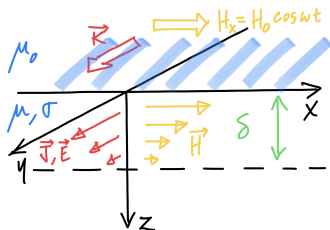
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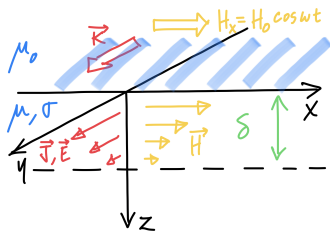
Skin depth, eddy currents, induction heating

- External magnetic field at the surface of conducting medium
- Spatially uniform
- Harmonic time dependence at frequency ω



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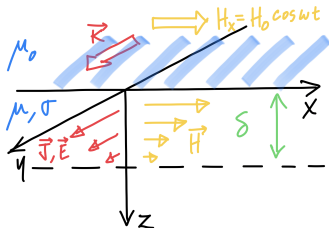
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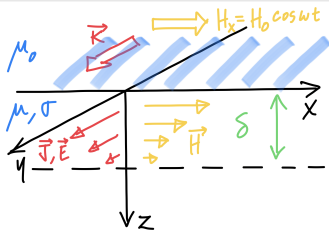
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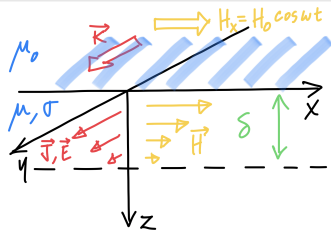
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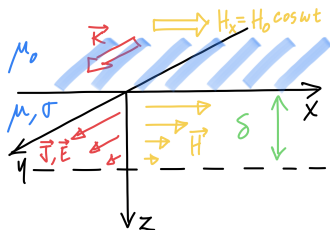
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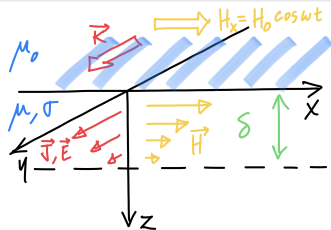
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$$E_y = \frac{\mu\omega\delta}{\sqrt{2}} H_0 e^{-z/\delta} \cos\left(\frac{z}{\delta} - \omega t + \frac{3\pi}{4}\right)$$

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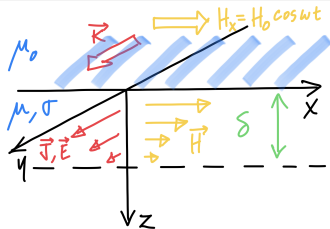
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- and corresponding current density

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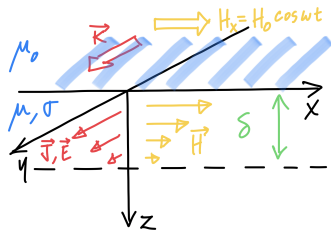
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$$K_y(t) = \int_0^\infty J_y(z, t) dz = -H_0 \cos \omega t$$

- Power dissipation confined to skin depth with density

$$P = \frac{\mu}{2} \omega H_0^2 e^{-2z/\delta}$$

Diffusion of magnetic field in conductors

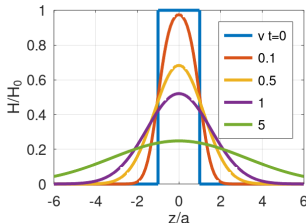
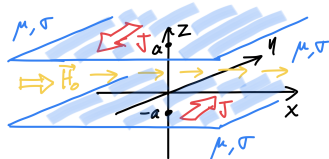
- Steady state uniform field $\vec{H}_0$ between current sheets

$$J_y = H_0 [\delta(z + a) - \delta(z - a)] [1 - \Theta(t)]$$

- $H(|z| > a) = 0$ for $t < 0$
- Currents turned off at $t = 0$
- Field diffuses into the bulk

$$H_x(z, t) = \frac{2}{\pi} H_0 \int_0^\infty d\kappa e^{-\nu t \kappa^2} \frac{\sin \kappa}{\kappa} \cos\left(\frac{z}{a} \kappa\right)$$

- with decay rate $\nu = (\mu\sigma a^2)^{-1}$



Sidebar on diffusion equation – heat transfer

- Consider the heat transfer eqn in a uniform isotropic medium

$$\nabla^2 T = \frac{1}{\alpha} \partial_t T + \frac{W}{k}$$

- describing the evolution of the temperature distribution $T(\vec{x}, t)$ with input power density $W(\vec{x}, t)$
- α is the *thermal diffusivity* which combines thermal conductivity k , specific heat c and density ρ

$$\alpha = \frac{k}{c\rho}, \quad \alpha_{Cu} \simeq 1.16 \text{ cm}^2/\text{s}$$

- The solution $G(x, t)$ for a spiked initial distribution $G(x, 0) = \delta(x)$ and no other input W

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- Motion of molecules (eg. ideal gas)
- described by number or charge density $\rho(\vec{x})$
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where D is the *diffusion coefficient*

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$$n \simeq \left[\frac{9.0 \text{ g/cm}^3}{64 \text{ g/mol}} \right] * \left[6.0 \times 10^{23} / \text{mol} \right] \simeq 8.5 \times 10^{22} / \text{cm}^3$$

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$$J = e n v_d = \sigma E$$

$$e \simeq 1.6 \times 10^{-19} \text{ C}, \quad \sigma \simeq 6.0 \times 10^7 / (\Omega \text{m})$$

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- The most probable speed from classical thermal considerations (Maxwell-Boltzmann distribution)

$$\frac{3}{2} kT \simeq 37 \text{ meV} = \frac{1}{2} m_e v_p^2 \rightarrow v_p \approx 100 \text{ km/s}$$

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- Speed from simple quantum free-electron gas model

$$\varepsilon_F = \frac{\hbar^2}{2 m_e} \left(3\pi^2 n \right)^{2/3} \simeq 7 \text{ eV} \rightarrow v_F \approx 1000 \text{ km/s}$$

What we've got here, is failure to communicate

How would you communicate with someone through a thick copper wall?

- a Flip on/off a (high-current) DC magnet
- b Flip on/off a (high-power) DC heater
- c Knock (speed of sound ~ 4.8 km/s)
- d Flip a switch wired to a battery and light bulb on the other side
- e ?

(1D) wave vs diffusion equation

$$\text{Wave eqn } \nabla^2 h = \frac{1}{c^2} \partial_t^2 h$$

$$\text{has general soln } h(x, t) = f(x \pm ct)$$

and represents a waveform $f(x)$ moving along $\pm x$ with speed c .
 $t \rightarrow -t$ changes the direction of the wave.

$$\text{Diffusion eqn } \nabla^2 h = \frac{1}{D} \partial_t h$$

$$\text{has soln } h(x, t) = \int dk f(k) e^{ikx} e^{-Dk^2 t}$$

representing a distortion of the initial shape

$$h_0(x) = h(x, 0) = \int dk f(k) e^{ikx}$$

where the small-scale (high- k) components dissipate faster.
 $t \rightarrow -t$ is not a solution.